Envelope encryption analysis

AWS Academy is the main lesson. This short deck connects it to our class work.

PRIMARY

AWS Academy

CSF Module 5 recap · local workshop

IN CLASS

Practice

Data-key and key-access reasoning

COURSE WORK

Local task

Local 08 · C5R1 Part 2 / C5R2 Part 1

Local 08 due 22 Feb · Use only your assigned Part

One path through the topic

STEP 1

Data key

Use a data key to encrypt the object data.

STEP 2

KMS key

Protect the data key with a KMS key and control its use.

STEP 3

Access

A later decrypt depends on policy and key state, not only stored ciphertext.

Why can ciphertext exist while decryption is unavailable?

What to do this week

BEFORE CLASS

Prepare

Bring the Week 15 KMS lab result and your assigned variant.

IN CLASS

Work

Trace encrypt, store and decrypt in the local scenario.

AFTER CLASS

Record

Submit your assigned Local 08 Part with a reasoned key-state check.

AWS work stays in AWS Academy · Local work stays in Classroom

SSE-KMS envelope and key states

UPLOAD

Two data-key copies

S3 requests a data key, uses the plaintext copy for object encryption, and keeps the encrypted copy with the object.

READ

KMS on later read

S3 discards the plaintext data key. In the worksheet's per-object SSE-KMS flow, a later read calls KMS to decrypt its stored copy.

KEY STATE

Recover or lose access

Disabled can be re-enabled; cancelled pending deletion leaves the key disabled; completed deletion is irreversible.

Local 08 uses per-object SSE-KMS without S3 Bucket Keys; ciphertext alone is not enough.